SEQUENCE 10

Variations and extrema of a function

1–12 February 2027

10.1 Increasing, decreasing and monotonicity

Key point · Definitions. On an interval I, f is increasing if for all $a, b \in I$, $a < b \Rightarrow f(a) \leq f(b)$; decreasing if $a < b \Rightarrow f(a) \geq f(b)$. The variations are strict when the inequalities between images are strict. A constant function has the same value everywhere; it is both increasing and decreasing in the broad sense. A function that is monotonic on I is increasing or decreasing there.

Method · Table of variations. Indicate the domain, the intervals of monotonicity, the values at the endpoints when defined, then the arrows. A sign table shows whether $f(x)$ is positive or negative; a table of variations shows how its values change.

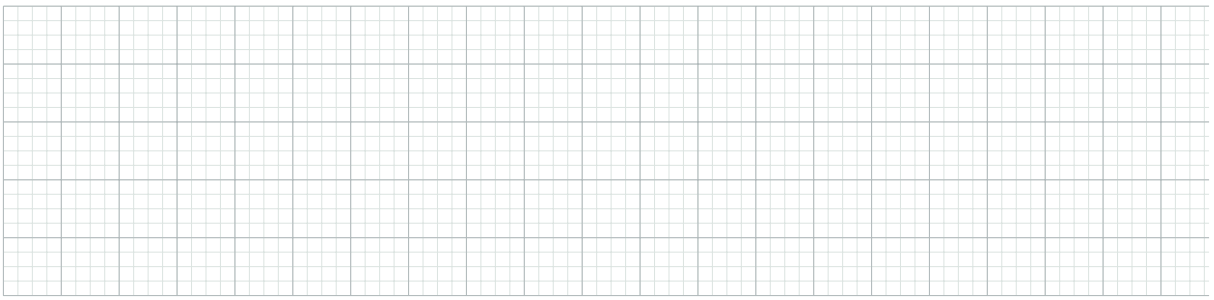
Example 1. On $[-3, 4]$, $f(x) = (x-1)^2 - 2$. Using the square, extending the curve studied in sequence 6, draw its table of variations. Give two numbers $a < b$ such that $f(a) > f(b)$.

Example 2. Can a function be negative and increasing? Give an example on an interval.

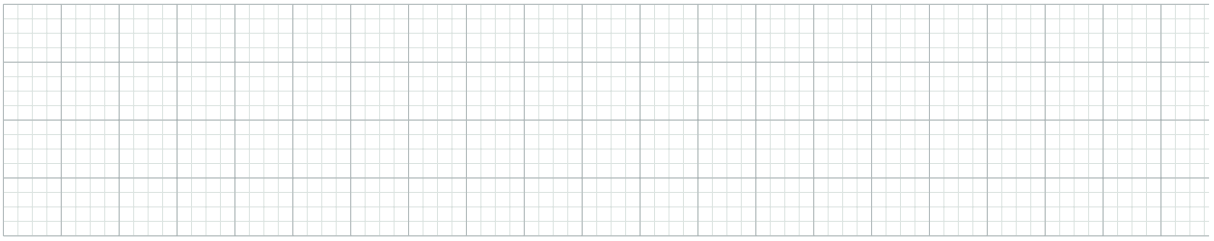
10.2 Maximum and minimum on an interval

Key point · Definitions. f has a maximum at $a \in I$ if, for every $x \in I$, $f(x) \leq f(a)$. The value of the maximum is $f(a)$. f has a minimum at $b \in I$ if, for every $x \in I$, $f(x) \geq f(b)$. An extremum must be attained. Distinguish its value from the abscissa where it is attained.

Example 3. Return to $f(x) = (x-1)^2 - 2$ on $[-3, 4]$. Give the minimum and maximum, then where they are attained. Repeat on $[0, 4]$.

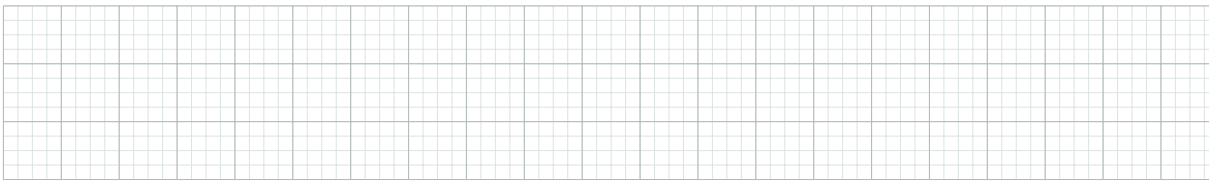


Example 4. Does the function $f(x) = x$ have a maximum on $[0, 1)$? A minimum?



Key point · Reading without inventing. A table of variations does not provide all the intermediate images. It allows comparing certain images; if the abscissae lie on different branches, the comparison may be indeterminate.

Example 5. Does the mere fact that a function decreases then increases allow comparing two images on either side of the minimum?



10.3 Variations of affine functions

Key point · Property and rate of change. For $f(x) = mx + p$ and $a \neq b$,

$$\frac{f(b) - f(a)}{b - a} = m.$$

If $m > 0$, f is strictly increasing; if $m < 0$, strictly decreasing; if $m = 0$, constant.

Proof to write.

Example 6. Determine the variations of $f(x) = 3x + 2$, $g(x) = 7 - 6x$, $h(x) = -x$. Give the sign of $2x - 6$ and explain why the sign of this function does not stay constant although it is increasing.

Example 7. An affine function u satisfies $u(2) = 5$, $u(6) = 13$. Determine u, then its minimum and maximum on $[-1, 4]$.

10.4 Variations of the square and the absolute value

Key point · Properties. $x \mapsto x^2$ and $x \mapsto |x|$ are strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0, +\infty)$. Their minimum on $\mathbb{R}$ is 0, attained only at 0.

Proof to write.

Example 8. Compare without a calculator: $(-3.2)^2$ and $(-3.1)^2$; 1.2^2 and 1.3^2; $|-5|$ and $|-2|$.

Example 9. Bound x^2 if $-3 \leq x \leq 2$. Why can we not simply square the two bounds?

10.5 Variations of the inverse function

Key point · Property. The inverse function is strictly decreasing on $(-\infty, 0)$ and on $(0, +\infty)$, separately. It is not decreasing on the whole of $\mathbb{R}^*$.

Proof to write.

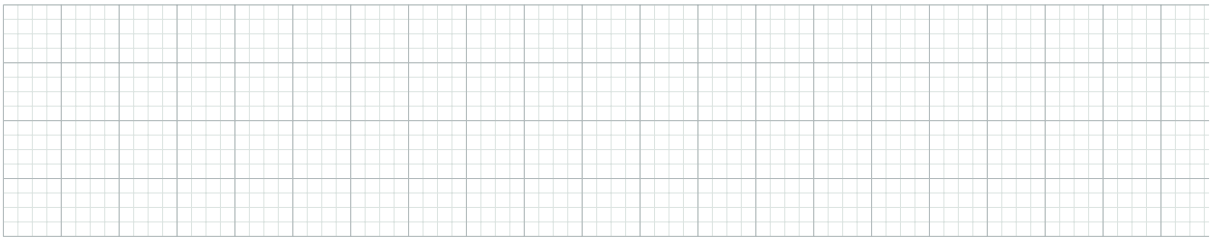
Example 10. Compare $1/(-4)$ and $1/(-2)$, then $1/3$ and $1/5$. Bound $1/x$ if $2 \leq x \leq 5$, then if $-5 \leq x \leq -2$.

Example 11. Why is the interval $[-1, 1]$ not suitable for directly applying a variation rule of the inverse function?

10.6 Relative position of a number and its square

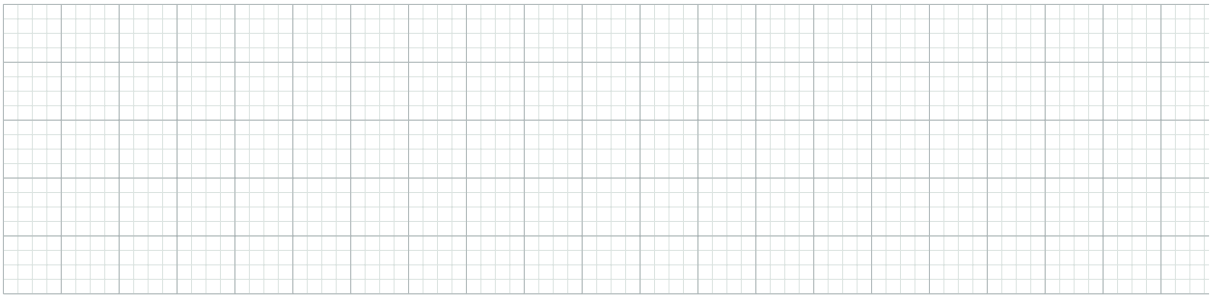
Key point · Property on the positives. For $0 \leq x \leq 1$, $x^2 \leq x$; for $x \geq 1$, $x^2 \geq x$. Equality holds at 0 and 1.

Proof to write.

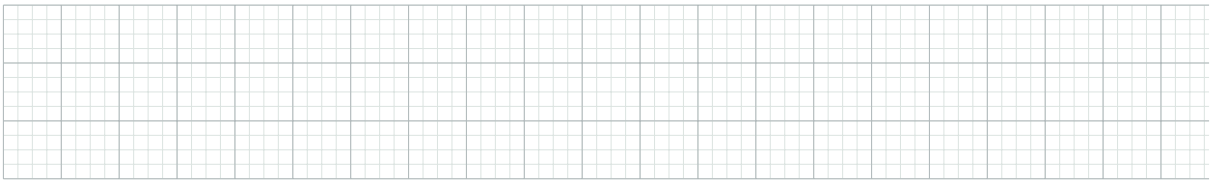


Key point · Square root and cube. The square root function is strictly increasing on $[0, +\infty)$. The cube function is strictly increasing on $\mathbb{R}$.

Example 12. Explain why the square root is increasing from the increase of the square on the positives. For the cube, use $b^3 - a^3 = (b-a)(a^2+ab+b^2)$.



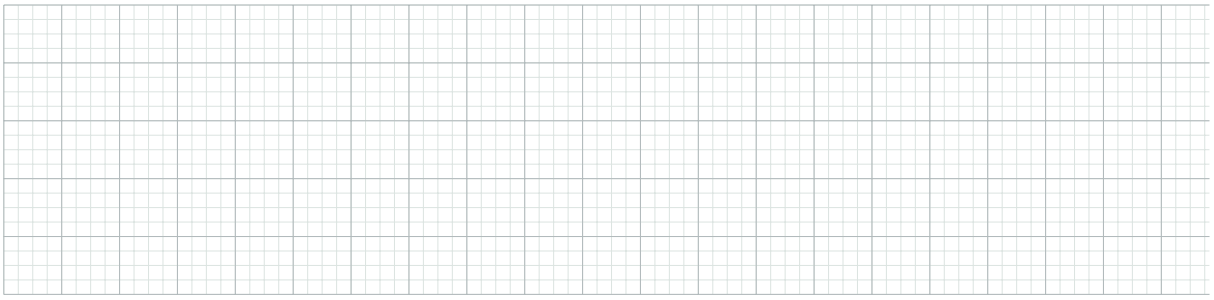
Example 13. Compare $\sqrt{5}$ and $\sqrt{6}$, then $(-2)^3$ and $(-1)^3$.



10.7 Optimising: conjecture and prove

Example 14. A rectangle of perimeter 20 m has area $A(x) = x(10 - x)$ with $x \in (0, 10)$.

1. Conjecture the maximum area using a table of values.
2. Show that $A(x) = 25 - (x - 5)^2$. Deduce the exact maximum and the corresponding dimensions.



```python
def aire(x):
    return x*(10-x)

def maximum_grille(N):
    # N entier >= 2 ; points strictement entre 0 et 10
    meilleur_x = 10/N
    meilleur_y = aire(meilleur_x)
    for k in range(2, N):
        x = 10*k/N
        if aire(x) > meilleur_y:
            meilleur_x, meilleur_y = x, aire(x)
    return meilleur_x, meilleur_y
```

Example 15. What does the program return for $N = 10$? Does a numerical sweep always prove the maximum of a function on an interval?

